

we have no control over c and cannot make any other assumptions about c other than it comes from the domain.

Existential instantiation: is the rule that allows us to conclude that there is an element c in the domain for which $P(c)$ is true if we know that $\exists x P(x)$ is true. We cannot select an arbitrary value of c here, but rather it must be a c for which $P(c)$ is true. Usually we have no knowledge of what c is, only that it exists. Because it exists, we may give it a name (c) and continue our argument.

Existential generalization: is the rule of inference that is used to conclude that $\exists x P(x)$ is true when a particular element c with $P(c)$ true is known. That is, if we know one element c in the domain for which $P(c)$ is true, then we know that $\exists x P(x)$ is true.

Table 2 Rules of Inference for Quantified Statements

Rule of Inference

Name

$\forall x P(x)$

Universal instantiation

$\therefore P(c)$

$P(c)$ for an arbitrary c

Universal generalization

$\therefore \forall x P(x)$

Existential instantiation

$\exists x P(x)$

$\therefore P(c)$ for some element c

Existential generalization

$P(c)$ for some element c

$\therefore \exists x P(x)$